Previous Year Question Papers

Subject: PD-PMD with CAD/CAM

1. The temperature at which the volume of a gas becomes zero is called
   a) absolute scale temperature b) absolute zero temperature
   c) absolute temperature d) none of these

2. The change of entropy, when heat is absorbed by the gas is
   a) positive b) negative c) both a & b d) none of these

3. In case of hyperbolic expansion of a gas, the heat supplied is __________________ the work done.
   a) equal to b) more than c) less than d) none of these

4. To which of the following are Maxwell's thermodynamics relations applicable ?
   a) Thermodynamic processes b) Mechanical System in equilibrium
   c) Chemical System in equilibrium d) Reversible process

5. Liquids have
   a) two distinct values of specific heat
   b) only one value of specific heat
   c) different values of specific heat at same temperature
   d) no specific heat

6. The percentage of carbon in crude oil varies between
   a) 83% to 87% b) 50% to 60% c) 90% to 94% d) 40% to 45%

7. The maximum stress in a ring under tension occurs
   a) along the line of action of load b) perpendicular to the line of action of load
   c) at 45° with the line of action of load d) none of these

8. The branch of engineering science, which deals with water at rest or in motion is called
   a) hydraulics b) fluid mechanics
   c) applied mechanics d) kinematics

9. A small hole in the side or base of a tank is termed as
   a) notch b) orifice c) mouthpiece d) downed orifice

10. Which of the following is an example of spherical pair ?
    a) ball and socket joint b) bolts and nut

c) ball bearing and roller bearing d) none of these

11. What is the unit of mass moment of inertia is S.I. units ?
a) Kg-m b) Kg-m2 c) m4 d) Nm/Kg

12. In a continuous system, the number of degree of freedom would be
a) one b) two c) three d) four

13. Under microscope, ferrite appears
a) white b) light c) dark d) none of these

14. Out of the following which is the amorphous material ?
a) Lead b) Brass c) Glass d) Silver

15. ________________ is not a ceramic material.
a) Glass b) Bakelite c) Clay d) Aluminium oxide

16. Efficiency energy of a substance depends on
a) volume b) pressure c) temperature d) entropy

17. Hydrometry deals with the
a) hydroscopic substances b) water vapour in air
c) temperature of air d) pressure of air

18. 1m3 of air at atmospheric condition weighs approximately
a) 0.5 kg b) 1.0 kg c) 1.3 kg d) 2.2 kg

19. A compressor at high altitude will draw
a) more power b) some power c) less power d) none of these

20. The unit of Stefan Boltzmann constant is
a) watt/cm2 °K b) watt/cm2 °K c) watt2/cm °K 4 d) watt/cm4 °K

21. The boiling point of ammonia is
a) -100°C b) -50°C c) -33°C d) 0°C

22. The specific weight of water is 1000 kg/m3
a) at normal pressure of 760 mm b) at 4°C temperature
c) at mean sea level d) all of these

23. A pair of Smith's Tong's is an example of the lever of
a) zeroth order b) first order c) second order d) third order

24. The total strain energy that can be stored in a body is known as
a) resilience b) proof resilience
c) modulus of resilience d) toughness

25. The periodic time of a simple pendulum depends on
a) size of bob b) mass of bob
c) material of bob d) length of pendulum

26. White cast iron contains carbon in the form of
a) free carbon b) graphite
c) cementite d) white carbon

27. Sphalerite is the principal ore or raw material for
a) zinc b) lead c) silver d) glass

28. Shear stress theory is applicable for
a) ductile materials b) brittle materials c) clastic materials d) all of these

29. One horsepower is equal to
a) 102 watts b) 75 watts c) 550 watts d) 735 watts

30. Which is not identical for an atom and an isotope
a) mass number b) atomic number
c) chemical properties d) all of these

31. The material used for coating the electrode is called
a) protective layer b) binder c) flux d) de-oxidiser

32. The work study is done by means of
a) planning chart b) process chart c) fixed expenses d) all of these

33. The purpose of micromotion study is to
a) help in collecting the motion time data for synthetic time standards
b) assist in finding out the most efficient way of doing work
c) train the individual operator regarding the motion economy principles
d) all the above

34. Standard time is equal to
a) normal time plus allowances b) normal time minus allowances
c) normal time taken by an operation d) none of these

35. Acetylene gas is generated from
a) carbon b) calcium carbide c) calcium d) calcium carbonate

36. The most commonly used flame in gas welding is
a) neutral b) oxidising c) carburising d) all of these

37. Oxygen to acetylene ratio in case of neutral flame is
a) 0.8 : 1.0 b) 1 : 1 c) 1.2 : 1 d) 2 : 1

38. CPM stands for
a) common planning method b) critical path method
c) critical process method d) combined process method

39. The type of layout suitable for automobile manufacturing concern is
a) product layout b) process layout
c) combination layout d) fixed position layout

40. The chart which gives an estimate about the amount of materials handling between various work stations is known as
a) process chart b) travel chart c) flow chart d) operation chart

41. Father of industry engineering is
a) Jeck gilberth b) Gantt c) Taylor d) Newton

42. Tick the odd man out
a) Taylor b) Drucker c) MC Gregar d) Galileo

43. PERT analysis is based on
a) optimistic time b) pessimistic time c) most likely time d) all of these

44. Bar charts are suitable for
a) minor works b) major works c) large projects d) all of these

45. The first free trade zone in india was established at
a) Kochi b) Goa c) Chennai d) Mumbai

46. Human resource planning includes
a) Raw material resources b) Recruitment and selection
c) Sales of the firm d) None of these

47. The basic objective of quality control in any organisation is

a) to build up customer good will
b) to ensure control
c) to achieve optimum cost
d) all of these

48. System cost includes the total amount for
a) Service life support
b) Development
c) Production
d) All of these

49. In a shop floor heavy jobs are lifted by means of
a) Fork lift
b) Conveyors
c) Hoists
d) Overhead crane

50. Why is job 'enrichment technique' applied ?
a) To make people happy
b) To reduce labour monotony
c) To overcome boring and demotivating work
d) All the above

51. Constant Volume Cycle is also known as
a) Otto cycle
b) Diesel cycle
c) Dual cycle
d) All of these

52. Which of the following has the two-stroke cycle engine ?
a) Auto cycles
b) Motor cycles
c) Scooters
d) All of these

53. Lanova combustion chamber is also known as
a) energy cell
b) open combustion chamber
c) pre-combustion chamber
d) squish combustion chamber

54. Balancing is one of the inspection factor in
a) crankshaft
b) piston
c) cylinder
d) crankcase

55. _____________ is the process by which the burnt gases escape from the cylinder.
a) Supercharging
b) Governing
c) Scavenging
d) Knocking

56. The choke is also called as
a) jet
b) venturi
c) strangler
d) float

57. How many cells are there in a 12 volt storage battery ?
a) 6
b) 9
c) 12
d) 3

58. The purpose of a suspension damper is to
a) resist the road shocks
b) reduce the bumb stroke of the spring
c) absorb the energy stored in the spring
d) none of these

59. The quantity of air-petrol mixture that enters the engine cylinder is regulated by the

a) throttle b) strangler c) float d) needle valve

60. The device used to reduce the exhaust noise is known as
a) exhaust manifold b) exhaust pipe c) tail pipe d) muffler

61. The Government of the English East India Company came to an end in
a) 1856 A.D b) 1857 A.D c) 1858 A.D d) 1862 A.D

62. ___________________ was the Viceroy responsible for the partition of Bengal
a) Lord Canning b) Lord Dalhousie c) Lord Curzon d) Lord Wellesley

63. Petrology deals with which of the following ?
a) Study of the petroleum products
b) Study of petroleum-related market economy
c) Study of rocks in the earth's crust
d) Study of the formation of soils

64. The Election Commission is constituted
a) every five years b) every ten years
c) every six years d) It is a permanent constitutional body

65. Which is the brightest planet as seen from the Earth ?
a) Mercury b) Pluto c) Jupiter d) Venus

66. In ______________ electrical energy is converted into light energy.
a) a battery b) an electric bulb c) a loudspeaker d) dynamo

67. The hardest substance known is
a) Diamond b) Steel c) Platinum d) Tungsten

68. The genetic material of a cell resides in
a) cytoplasm b) protoplasm c) ribosome d) DNA

69. Which of the following organism does not contain chlorophyll ?
a) Mosses b) Ferns c) Algae d) Fungi

70. Which one of the following birds can twist its head in a complete circle ?
a) Owl b) Crow c) Peacock d) None of these

71. Which material can't be processed in compression moulding
a) PF b) UF c) MF d) PET

72. The parting line location is determined by the
a) Shape of the mould b) Shape of the component

c) Size of the part d) Press Tonnage

73. The clip on a pen is an example of
a) under feed mould b) two plate mould
c) split mould d) none

74. Curing means
a) Material solidify under heat & pressure b) Material degrade
c) Material injected during process d) Material cooled & become solid

75. Deflashing is
a) Used for removing additional material b) Is always a part of finishing operation
c) Both a & b d) None of above

76. Which of the following polymers is self extinguishing in nature ?
a) PVC b) CPVC c) PVDC d) All of these

77. Preheating of material in transfer moulding is
a) Necessary b) Often used c) Not necessary d) No meaning

78. Which form of PMMA is widely available in the market ?
a) Solution b) Powder c) Bead d) Granules

79. Venting in mould depends upon
a) Plastic compounds viscosity b) Heating pattern of the mould
c) Both of above d) None of above

80. Thermoforming process is a __________________ processing technique.
a) Primary b) Secondary c) Tertiary d) None of these

81. While processing Nylon 66 it is important to control drooling
a) By cooling nozzle b) By nozzle of reverse type
c) By lowering hopper temperature d) By lowering cylinder temperature

82. Matched die Forming process is similar to
a) Calendering b) Injection Moulding
c) Extrusion d) Compression Moulding

83. The suitable material for X-ray film can be made from
a) PET b) Nylon 66 c) Polyamide d) PBT

84. How does the viscosity of PVC measured ?
a) MFI b) Capilary c) K-value d) None of these

85. Name the plasticizer commonly used in PVC.
a) Solvent b) DOP c) Silicon oil d) All of these

86. Name the blender used to premix the PVC resin
a) Mixer b) Ribbon blender c) Two plate mixer d) None of these

87. About 50 to 60% composite manufacturing industry uses
a) Epoxy Resin b) Furan
c) Vinyl ester resin d) Unsaturated polyester

88. The function of an extruder is not to
a) Convert a solid polymer feed stock to a homogeneous melt
b) Pump it and deliver it at a constant rate and pressure
c) Maintain uniform temperature
d) Give shape to product

89. Which of the following combination is an essential part of a composite ?
a) Filler and pigments b) Catalyst and lubricants
c) Plastic resin and reinforcement d) Resin and stabilizer

90. Find out the crystalline material from the following
a) PS b) PMMA c) SAN d) HDPE

91. PET bottle is manufactured by
a) Stretch Blow Moulding b) Blow Moulding
c) Injection Moulding d) Extrusion Process

92. In extrusion process the function of screen pack is ______________________.
a) To filter contamination b) Arrest unmelted particles
c) Developing backpressure d) All of these

93. Which of the following is having lowest density ?
a) LLDPE b) LDPE c) PP d) PS

94. Choose from the following suitable composite plastic for making boat hull at low cost
a) Unsaturated polyester b) Polyurethane
c) Epoxy resin d) Phenolic plastic

95. If self threading screws is to be made of engineering plastics then
a) Polyacetal is the best b) Nylon 6 is the best

c) PBT is the best
d) PET is the best

96. Choose the correct FRP technique for making pipe with large diameter.
a) Pultrusion
b) Continuous laminate
c) Centrifugal casting
d) Filament winding

97. Out of the following materials which has the highest density
a) LDPE
b) LLDPE
c) UHMHDPE
d) HDPE

98. Which is the following material is transparent in nature ?
a) POM
b) PVC
c) Nylon
d) ABS

99. Application of PC in precision optics has been found due to
a) Optical property and toughness
b) Optical property and high viscosity
c) Optical property and high temperature stability
d) Optical property and flammability

100. The ability of PBT to perform at high temperature is illustrated by
a) HDT
b) Flammability
c) Smoke Density
d) Ignition Test

101. Percentage of chromium in 18 – 4 – 1 high speed steel is
a) 18
b) 9
c) 4
d) 1

102. Duralumin contains
a) Aluminium, copper, manganese and magnesium
b) Aluminium, zinc and magnesium
c) Aluminium, tin and zinc
d) Aluminium, zinc, lead

103. Newton is the force which gives a body of 1 kg of mass an acceleration of
a) 1 cm/sec^2
b) 1 m/sec^2
c) 10 cm/sec^2
d) 980 cm/sec^2

104. A stationary object can have
a) K.E.
b) P.E.
c) Both a & b
d) Momentum

105. If x is between 0 and 1, which of the following increases as x increases.
I. $1-x^2$
II. $x-1$
III. $\frac{1}{x^2}$
a) I only
b) II only
c) III only
d) I and II only

106. If pencils are bought at 34 cents per dozen and sold at 3 for 10 cents, the total profit on $5\frac{1}{2}$ dozen is
   a) 24 b) 28 c) 30 d) 33

107. Steel is a
   a) Interstitial solid solution of carbon in iron
   b) Substitutional solid solution of iron and carbon in iron and cementite
   c) Interstitial solid solution of carbon in iron and cementite
   d) Substitutional solid solution iron and martensite

108. Which has highest specific gravity in the following?
   a) Aluminium b) Copper c) Silver d) Gold

109. The shaft of a motor rotates at a constant angular velocity of 3000 R.P.M. The radians it has turned through in 1 second is
   a) 1000π b) 100π c) 10π d) 5π

110. 35 is 7% of
   a) 5 b) 50 c) 500 d) 5000

111. The work done in an adiabatic change on a particular gas depends upon only
   a) Change in value b) Change in pressure
   c) Change in temperature d) None of these

112. The first law of thermodynamics is concerned with the conservation of
   a) Number of molecules b) Temperature
   c) Internal energy d) Number of moles

113. Air in a cylinder is suddenly compressed by a piston with the passage of time
   a) The pressure decreases
   b) The pressure increases
   c) The pressure may remain constant
   d) The pressure may increase or decrease depending upon the nature of gas

114. Diffusion is
   a) Mixing of unlike fluids
   b) Mixing of two portions of fluid
   c) Mixing of a gas in two containers at different pressure
   d) Mixing of two portions of a fluid at different temperature

115. The kerosene is generally used as a fuel
   a) For road fraction b) For lighting and cooking
   c) In thermal power plants d) For industrial furnaces

116. Which of the following is a scalar ?
a) Linear momentum b) Electric current
c) Weight d) None of these

117. If a body moves on a circular path with uniform speed, the acceleration of the body
a) Remains constant b) Changes
c) Acts away from the centre d) Is zero

118. A man can throw a ball up to a maximum distance x metres on a horizontal plane. The maximum height upto which he can throw the ball is
a) $\frac{x}{2}$ metres b) x metres c) $2x$ metres d) None of these

119. Which one of the following is more elastic ?
a) Rubber b) Steel c) Aluminium d) Glass

120. A perfect rigid body has value of Young's modulus
a) Zero b) 1 c) Infinite d) None of these

121. Simple stress is often called
a) Direct stress b) Transverse stress
c) Total stress d) None of these

122. If l and δl are the length and change in length respectively the strain is equal to
a) $\frac{\delta l}{l}$ b) $\frac{l}{l\delta}$ c) $l \times \delta l$ d) None of these

123. When compared to a riveted joint a welded joint has ___________ strength.
a) lesser b) greater c) Either (a) or (b) d) None of these

124. _______________ is a process of joining two pieces of metal by fusion.
a) Riveting b) Welding c) Either (a) or (b) d) None of these

125. Vessels used for storing fluid under pressure are called ______________.
a) Cylinders b) Spheres c) Shells d) None of these

126. The moment of inertia about a principal axis is called
a) Mass moment of inertia b) Second moment of inertia
c) Principal moment of inertia d) Any one of these

127. The impact strength of a material is an index of its ________________.
a) Resistance of corrosion b) Hardness
c) Toughness d) None of these

128. The ratio of lateral strain to linear strain is known as
a) Modulus of elasticity b) Modulus of rigidity
c) Poisson's ratio d) Elastic limit

129. In S.I system the unit of torque is _______________.
a) Kg. m b) Kg / cm^2 c) Newton metre d) Dynes

130. A rivet joint may fail due to ____________________.
a) Tearing of the plate b) Shearing of rivets
c) Crushing of rivets d) Any one of these

131. ______________ possess no definite volume and is compressible.
a) Solid b) Liquid c) Gas d) Vapour

132. The pressure ______________ as the depth of the liquid increases.
a) Increases b) Decreases
c) Remains unchanged d) None of these

133. Poise is a unit of _______________
a) Surface tension b) Viscosity
c) Specific weight d) Pressure

134. The force per unit area is called ________________.
a) Pressure b) Strain c) Surface tension d) None of these

135. The discharge through a pipe can be measured with ___________________.
a) A venturimeter b) An orificameter c) A flow nozzle d) All of these

136. The stability of a floating body depends upon _________________.
a) its volume b) its weight
c) its metacentric height d) the specific weight of fluid

137. The velocity ratio of the belt drive due to slip of the belt _____________.
a) increases b) decreases
c) remains unchanged d) unpredictable

138. Which of the following brakes is used in motor cars ?
a) Band brake b) Internal expanding brake
c) Shoe brake d) Any one of these

139. ________________ drive is not a positive drive.
a) V-belt b) Rope c) Flat-belt d) All of these

140. A universal joint is an example of _____________ pair.

a) Sliding b) Lower c) Higher d) None of these

141. What is the contact ratio for gears ?
a) Less than one
b) Zero
c) Greater than one
d) None of these

142. _____________ test is a non-destructive test.
a) Impact b) Charpy c) Radiography d) Tensile

143. Nickel is ________________ material.
a) Dielectric b) Ferro-electric c) Ferro-magnetic d) Dia-magnetic

144. ___________________ structure can be studied by naked eye.
a) Atomic b) Grain c) Micro d) Macro

145. _____________ will exhibit viscoelastic behaviour.
a) Steel
b) Diamond
c) Organic polymers
d) Neoprene

146. The following is an S.I. engine.
a) Diesel engine b) Petrol engine c) Gas engine d) None of these

147. Compression ratio of petrol engines is in the range of
a) 2 to 3 b) 7 to 10 c) 16 to 20 d) 80 to 90

148. Carburettor is used for ______________.
a) S.I. engine b) Gas engine c) C.I. engine d) None of these

149. ________________ is used for the insulating body of a spark plug.
a) Dolomite b) Alumina c) Glass d) Silica

150. ____________ lubrication technique is used for lubrication of the cylinder of a scooter engine.
a) Petrol b) Splash c) Gravity feed d) Forced feed

151. Crank shafts are generally ______________.
a) Die cast
b) Sand cast
c) Forged
d) Turned from bar stock

152. Which one is not related to I.C. engine ?
a) Gas turbine
b) 4 - stroke C.I. engine
c) Steam turbine
d) None of these

153. In a six cylinder I.C engine the number of spark plugs required are
a) 6 b) 1 c) 12 d) 0

154. In four stroke four cylinder I.C engine the number of spark plugs used are
a) Four b) Eight c) One d) Zero

155. Visco meter is the instrument used for finding out the fluids.
a) Flash point b) Viscosity c) Fire point d) None of these

156. Basic two plate moulds have ___________ daylight.
a) single b) double c) triple d) none of these

157. Draft provided in plastic parts for _______________________.
a) easy ejection b) tough ejection c) both d) none of these

158. ______________ reduces stress concentration and streamline the melt flow.
a) Cylindrical b) Sharp corner c) Fillet d) None of these

159. Dimensional difference between the mould and the moulding is ____________.
a) sharp corner b) shrinkage c) fillets d) none of these

160. Plastic part wall thickness should be ______ to eliminate internal stresses, warpage.
a) non-uniform b) uniform c) rough d) all of these

161. Expand the name PPO ?
a) Poly Propylene oxide b) Poly phenylene oxide
c) Poly Phenol oxide d) Poly phenylene oxygen

162. ____________material is used to manufacturing pipes for irrigation.
a) PF b) PVC c) LLDPE d) PS

163. Carbonated soft drink containers are made from ___________________.
a) POM b) Nylon 66 c) PET d) PPO

164. ________________ is a Thermoset Material.
a) MF b) PP c) PET d) None of these

165. The specific gravity of polyethylene is _____________ than that of water.
a) Higher b) Less c) Equal d) None of these

166. Jerry can with handle is produced by _____________.
a) Injection blow moulding b) Extrusion blow moulding
c) Injection stretch blow moulding d) Extrusion stretch blow moulding

167. In thermoforming plastic sheet is heated to the
a) Melting point b) Sag point c) Tg –point d) none of these

168. The term "Cull" is related to ____________ process.
a) Injection b) Compression c) Transfer d) Extrusion

169. Helix angle of the screw is ____________________.
a) 18° b) 17.8° c) 45° d) 25°

170. Bulk factor is the ratio of ______________________.
a) Volume of the Loose powder to the product
b) Volume of the product to the Loose powder
c) Area of the Loose powder to the product
d) None of the above

171. Which of the following numbers is the largest ?
a) $(2+2+2)^2$ b) $\{(2+2)^2\}^2$ c) $(2\times 2\times 2)^2$ d) $2+2^2+(2)^2$

172. In a Camp there is provision for 1600 participants for 60 days. Actually 1200 participated. How many days will the provision last ?
a) 75 b) 90 c) 80 d) 100

173. 8 men can build a foundation 18 metres long, 2 m broad and 12 m high in 10 days working 9 hours a day. Find how many men will be able to build a foundation 32 m long, 3 m broad and 9 m high in 8 days working 6 hours ?
a) 20 men b) 10 men c) 15 men d) 30 men

174. The area of a square is 225 square metres. Then the perimeter of the square is
a) 50 b) 15 c) $15\sqrt{2}$ d) 60

175. A : B = 4 : 5, B : C = 3 : 8; A : C = ?
a) 4 : 5 b) 12 : 40 c) 8 : 5 d) 5 : 8

176. I told him to pick up the newspaper which ____________ on the table.
a) lay b) laid c) lied d) none of these

177. Can you pay ________ all these articles ?
a) for b) of c) off d) out

178. It is time that we ____________ something useful.
a) may do b) did

c) should have been done d) could have been done

179. Indifferent means
a) antipathetic b) pathetic c) apathetic d) none of these

180. Warren means
a) tenement b) judicial order c) small bird d) gully

181. Boom means
a) slump b) prosper c) fall d) increase

182. Gandhiji played a ___________ (having a definite result) role in India's liberation.
a) demarcating b) delightful c) decisive d) devout

183. It is highly _____________ novel (impressive in language).
a) rhyming b) retoric c) rhetoric d) rhetorical

184. We can live without our friends but not without our __________________.
a) family b) necessaries c) pleasures d) neighbour

185. Same people want to become rich by _____________.
a) all means b) hook or crook c) no means d) hard work

186. If A = 1, B = 3, C = 5 and so on what do the numbers 3, 9, 7 stand for ?
a) BID b) BAD c) BED d) CAR

187. Find the odd man out
a) Christmas b) Diwali c) Holi d) Tuesday

188. Insert the missing number : 514, 258, 130, 66, …….
a) 34 b) 33 c) 36 d) 38

189. Woman is to womanly as child is _______________.
a) childish b) childlike c) childly d) childwise

190. Deer is to fawn as cow is to ______________.
a) calf b) elver c) cygnet d) foal

191. The purpose of introducing uniform for the Children in Schools is
a) for attention
b) for creating equality feeling
c) for fossring unity
d) for identification of the institution

192. Name the income of the money invested in education by Indian Government.
a) Taxes
b) Services of the Professionals
c) Educational cess
d) Political Donations

193. A test of "Shiftability" of base is called
a) Timer reversal test
b) Factor reversal test
c) Circular test
d) None of these

194. Which one of the following excretory organ ?
a) lungs
b) kidney
c) skin
d) small intestine

195. Decible is a unit of
a) heat
b) light
c) intensity of sound
d) none of these

196. For respiration, fish uses
a) gills
b) mouth
c) body
d) tail

197. The earliest of the vedas is
a) Rig
b) Yajur
c) Sama
d) Atharva

198. The river Ganga is divided into two branches
a) Padma & Bhagirathi
b) Chenab & Ravi
c) Sutlej & Beas
d) Ravi & Yamuna

199. The Ajanta Pantings belong to the
a) Buddhist period
b) Gupta period
c) Harappan period
d) Mauryan period

200. Which state is the largest producer of tea in India ?
a) West Bengal
b) Kerala
c) Assam
d) Tamil Nadu

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